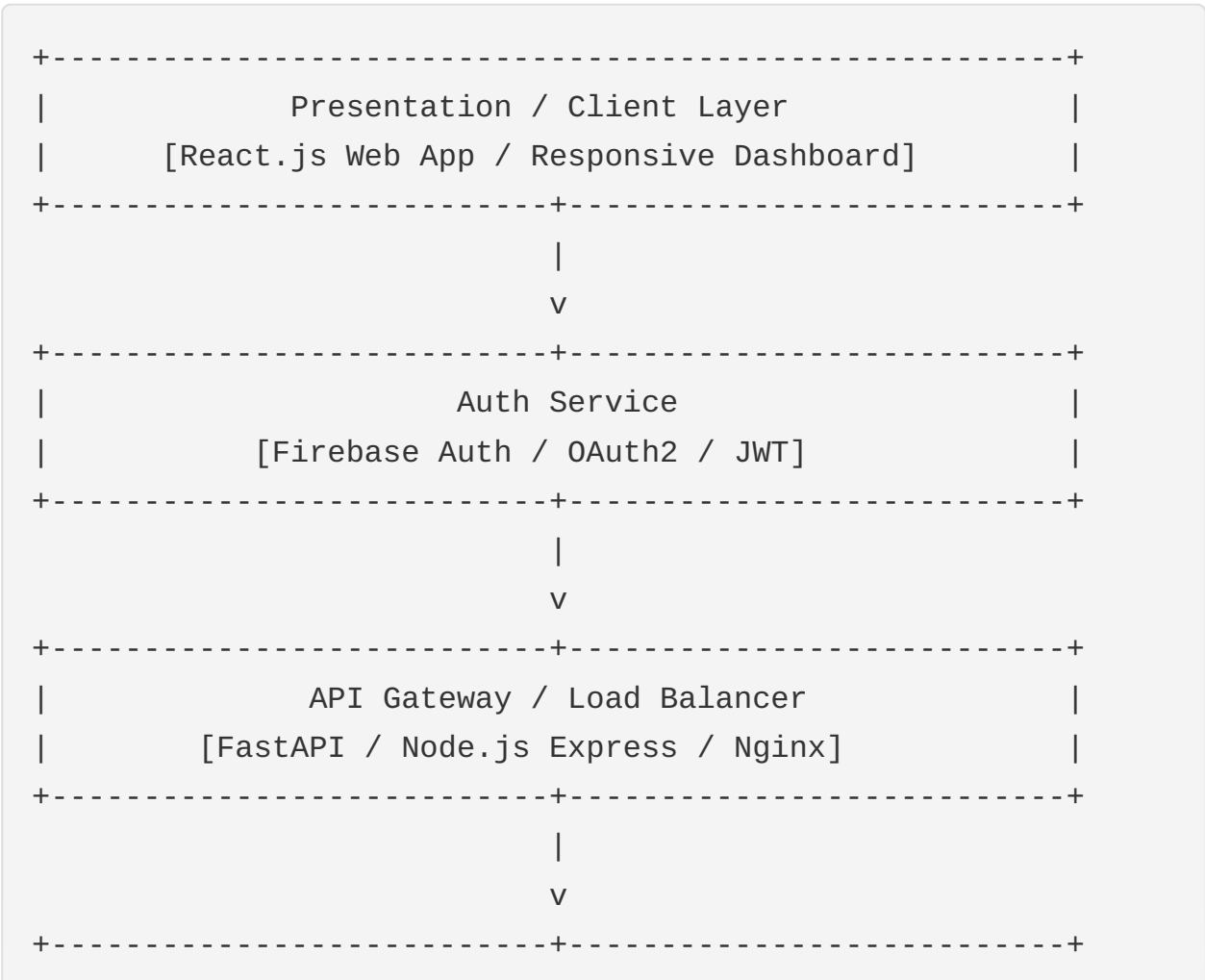


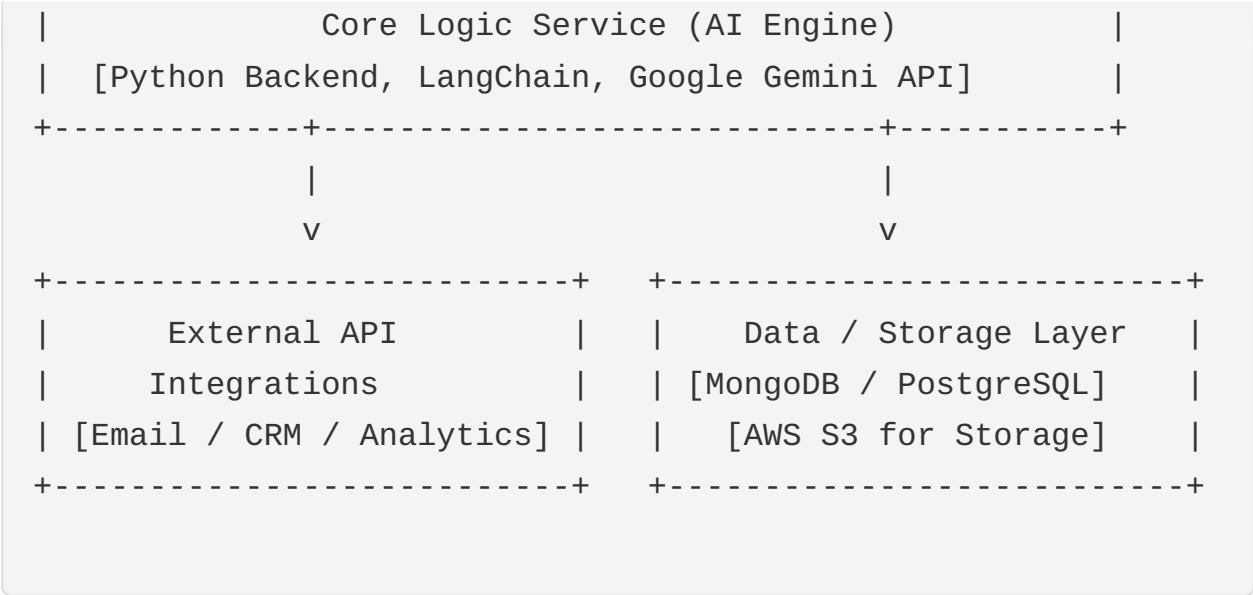
SMARTBRIDGE

Let's Bridge the Gap

Date	1 July 2026
Team ID	XXXX
Project Name	Customer Persona Generator - AI Powered Marketing Persona Application
Maximum Marks	5 Marks

Solution Architecture Diagram





Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	The user interface where marketing professionals input target audience data and view generated personas.	React.js, Tailwind CSS, Framer Motion
API Gateway	Handles incoming traffic, authenticates user sessions, and routes requests to specific microservices.	FastAPI, Nginx, JWT
Microservices (AI Engine)	The core logic component that processes demographic inputs and interacts with LLMs to generate persona insights.	Python, LangChain, Google Gemini API
Database	Stores user profiles, historical persona data, and marketing campaign configurations.	MongoDB (NoSQL), AWS S3 (for storage)